

Muhammad Hamza

Researcher in Quantum Information Science, Quantum Computing for Communication Systems, and Artificial Intelligence, with experience in reinforcement learning, hybrid quantum–classical learning, large language models, and intelligent communication systems.

Graduate Research Assistant at CQILAB¹
Kyung Hee University, South Korea

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Education

- **Kyung Hee University, Yongin-si, Korea** **Mar. 2025 – Feb. 2027 (Expected)**
M.S. in Electronics and Information Convergence Engineering
Advisor: Prof. Hyundong Shin
Research Area: Reinforcement Learning based Quantum Circuit Optimization and Hybrid Quantum–Classical Learning for 6G Communication Systems
- **National University of Sciences and Technology (NUST), Pakistan** **Oct. 2020 – Jun. 2024**
Bachelors in Electrical Engineering
Final Year Project: *Violence Detection using Deep Learning and ResNet-50*

Research Interests

My research interests span quantum information science, quantum computing, hybrid quantum–classical machine learning, and intelligent communication systems. I am particularly interested in applying reinforcement learning, deep learning, large language models, and agentic AI to problems involving quantum circuit design, communication-system intelligence, and autonomous decision making. My broader research goals include the development, optimization, and benchmarking of reliable, resource efficient, and deployable quantum and AI frameworks for next-generation communication, computing, and sensing systems.

Technical Skills

Quantum Circuit Optimization, Variational Quantum Circuits, Neural Networks, Hybrid Quantum–Classical Machine Learning, Large Language Models, Reinforcement Learning, Deep Learning, Agentic AI, Python, PyTorch, TensorFlow, Scikit-learn, Qiskit, PennyLane, GPU/CUDA Programming, LaTeX.

Honors and Awards

- **MS Research Scholarship, BrainKorea21 Plus** **2025 – 2027**
- **Graduate Student Scholarship, Kyung Hee University** **2025 – 2027**

¹Communications and Quantum Information Laboratory (CQILAB)

Experience

Work Experience

- **Kyung Hee University, Yongin-si, Korea** **Mar. 2025 – Feb. 2027 (Expected)**
Graduate Research Assistant — CQILAB
Department of Electronics and Information Convergence Engineering
 - Conduct research in quantum information science, quantum machine learning, reinforcement learning, and intelligent communication systems.
 - Developed reinforcement learning based methods for adaptive quantum circuit pruning, targeting gate complexity reduction while preserving quantum state fidelity.
 - Implemented and evaluated hybrid quantum-classical and classical machine learning models using Qiskit, PyTorch, TensorFlow, and Python.
 - Developed AI based decision frameworks for wireless communication systems using beam power sequences, convolutional neural networks, object detection models, and historical link state information.
 - Investigated the use of large language models and few shot prompting for sequence based blockage classification and compared their performance with conventional deep learning models.
 - Conduct literature reviews and present research seminars on quantum information, quantum machine learning, artificial intelligence, and next generation communication systems.
 - Contribute to research manuscripts, experimental design, performance benchmarking, technical visualization, and reproducible research workflows.
- **CMPak Zong, Pakistan** **Oct. 2024 - Mar. 2025**
Project Management Operations and Software Intern / Engineer
Telecommunications Sector
 - Worked with the Project Management Operations team on telecom project coordination, documentation, tracking, and operational support.
 - Supported software related tasks involving PostgreSQL database management, data handling, and backend system maintenance.
 - Gained practical experience in telecom operations, project workflows, database systems, and cross-functional engineering collaboration.

Research Highlights

- **Reinforcement Learning-based Quantum Circuit Pruning:** Developed an RL based framework for adaptive pruning of variational quantum circuits. The method aims to reduce gate complexity while preserving fidelity and model performance.
- **Agentic AI for RIS/LOS Communication Switching:** Developed an intelligent decision framework for RIS/LOS switching under blockage conditions in mmWave communication systems using beam power sequences, CNN based prediction, YOLO based vehicle detection, and previous link state information.

Publications & Presentations

Refereed Conference Proceedings

- [1] M. Hamza, M. M. U. Gondel, and H. Shin, “Adaptive Quantum Circuit Pruning with Reinforcement Learning,” in *Proc. Korea Information and Communications Society (KICS) Conference*,

Korea, 2025.

- [2] M. M. U. Gondel, M. Hamza, and H. Shin, “Agentic AI-based LOS/RIS Switching for mmWave Communication under Blockage Conditions,” manuscript in preparation.

Certifications

- **Certified Kubernetes Application Developer (CKAD) and Certified Kubernetes Administrator (CKA)**
The Linux Foundation **Mar. 2024 - May. 2024**